



Venkatasai Mannava

M.Tech Communications & Signal Processing - RA

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Degree	University/Institute	Year	CGPA/ Marks(%)
M.Tech Communications & Signal Processing - RA	IIT Hyderabad	2027	8.55
B.Tech in Electronics and Communication Engineering	Vignan's Foundation for science technology and research university	2023	9.61
XII (Board of Intermediate Education, Andhra Pradesh)	Vignan Cooperative junior college	2019	10.00
X (Central Board of secondary education)	Sri Somanatha Green field public school	2017	10.00

ABOUT

Communications and Signal Processing Engineer with hands-on experience in real-time GNSS receiver development. Currently pursuing an M.Tech at IIT Hyderabad, with practical experience in implementing GPS L1 C/A baseband signal-processing algorithms on DSP platforms. Proficient in C, Python, MATLAB, and SDR-based signal processing, with a focus on the development, optimisation, and real-time implementation of communication and signal-processing systems.

TECHNICAL SKILLS

- **Programming Languages** : C, Verilog, Python, MATLAB
- **Embedded DSP** : Implementation and optimization of real-time signal processing algorithms using fixed-point arithmetic, with emphasis on computational efficiency, numerical precision, and memory-constrained embedded DSP architectures.
- **Low-Level Debugging** : Debugging C programs on target hardware through register-level inspection.
- **Memory Mapping** : Developing linker scripts for custom memory mapping and efficient memory utilization.
- **RF Test Equipment** : Hands-on experience with Rohde & Schwarz spectrum analyzers and signal generators for signal characterisation and power measurements.
- **Software-Defined Radio** : Hands-on experience with SDR platforms for real-time signal transmission, reception, and I/Q data processing.
- **Scripting and Automation** : Proficient in Python and Bash scripting for test automation, data processing, and Monte Carlo simulation for baseband processing blocks.
- **Development Tools** : Linux, GCC, GDB, Make, Git, VS Code, Bash

EXPERIENCE

5G Testbed R & D Lab, IIT Hyderabad
M.Tech RA

Aug 2024 - Present

- Ported and optimised the **GPS L1 C/A base band receiver** for a resource-constrained **CEVA DSP** using fixed-point arithmetic in C, targeting a platform with a **153 MHz clock frequency and 512 KB on chip memory (code and data memory)**.
- Optimised the receiver signal-processing pipeline to achieve a **tracking execution time of 160 us per satellite**, enabling efficient real-time multi-channel processing within the available computational resources.
- Performed extensive **Monte Carlo testing** across different geographic locations and signal conditions, demonstrating positioning performance comparable to the floating-point implementation.
- Conducted mathematical analysis of the complete receiver signal-processing chain and characterised the **noise performance and sensitivity** of individual processing stages.
- Achieved a receiver sensitivity of approximately **-140 dBm for acquisition** and **-150 dBm for tracking**, with a **median positioning accuracy of approximately 7 m**.
- Identified, analysed, and resolved complex **algorithmic and implementation-level challenges** during DSP integration, addressing fixed-point precision, computational complexity, memory utilisation, and real-time processing constraints.

5G Testbed R & D Lab, IIT Hyderabad
Project Associate

Feb 2024 - July 2024

- Developed a strong understanding of the GPS L1 C/A signal structure and receiver signal-processing algorithms.
- Designed and implemented a complete GPS L1 C/A base band receiver in C using floating-point arithmetic, comprising acquisition, tracking, bit synchronisation, frame synchronisation, and PVT estimation modules.
- Validated the receiver using live GPS L1 C/A signals captured with a Blade RF SDR and evaluated receiver performance through extensive Monte Carlo simulations.
- Achieved a median positioning accuracy of approximately 7 m while tracking signals from six satellites.

PROJECTS

GPS L1 C/A Receiver on DSP

- As part of my M.Tech thesis, developed and optimised a **real-time GPS L1 C/A base band receiver on a CEVA DSP** using fixed-point arithmetic for integration into an **NB-IoT chip set** targeting navigation and NTN applications. The receiver achieved **-140 dBm acquisition sensitivity**, **-150 dBm tracking sensitivity**, and a **median positioning accuracy of approximately 7 m**, demonstrating its suitability for deployment in commercial GNSS-enabled applications.

Laboratory Assignments

- Implemented a **fixed-point FIR filter** in **Verilog**.
- Implemented **Newton-Raphson-based division** in **Verilog** using fixed-point arithmetic.
- Simulated **digital modulation schemes** and analysed BER over **AWGN and Rayleigh fading channels** using Python.
- Simulated **MIMO systems** with different **Tx/Rx configurations** and analysed BER performance using Python.

- Implemented **OFDM over AWGN and Rayleigh fading channels** using Python.

RELEVANT COURSES

- **Random Variables and Stochastic Processes** : Statistical modelling, CDF analysis, and receiver sensitivity estimation.
- **Principles of Digital Communications** : BPSK modulation/demodulation, BER analysis, and maximum-likelihood detection.
- **Advanced Digital Signal Processing** : Sampling theory, digital filtering, and up sampling/down sampling techniques.
- **Wireless Communications** : Doppler, carrier phase, and code phase estimation using maximum-likelihood techniques.
- **Matrix Theory** : Nonlinear equation solving and least-squares estimation for GPS PVT computation.

POSITIONS OF RESPONSIBILITY

- **TA for FPGA & DSP lab** : Guided undergrad students in python,verilog and fixed point arithmetic.
- **Student Coordinator** : CIRH International Conference.